Computer Vision

Instructor: Luigi Cinque
cinque@di.uniroma1.it

HISTOGRAM TRANSFORMATION IN IMAGE PROCESSING AND ITS APPLICATIONS

Contents

- Histogram
- Histogram transformation
- Histogram equalization
- Contrast streching
- Applications

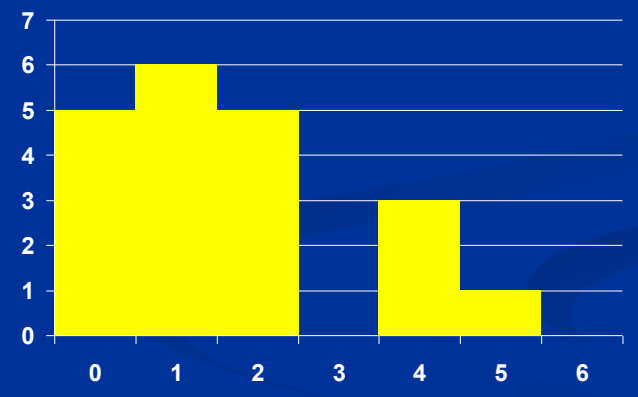
Histogram

The (intensity or brightness) histogram shows how many times a particular grey level (intensity) appears in an image.

For example, 0 - black, 255 – white

0	1	1	2	4
2	1	0	0	2
5	2	0	0	4
1	1	2	4	1

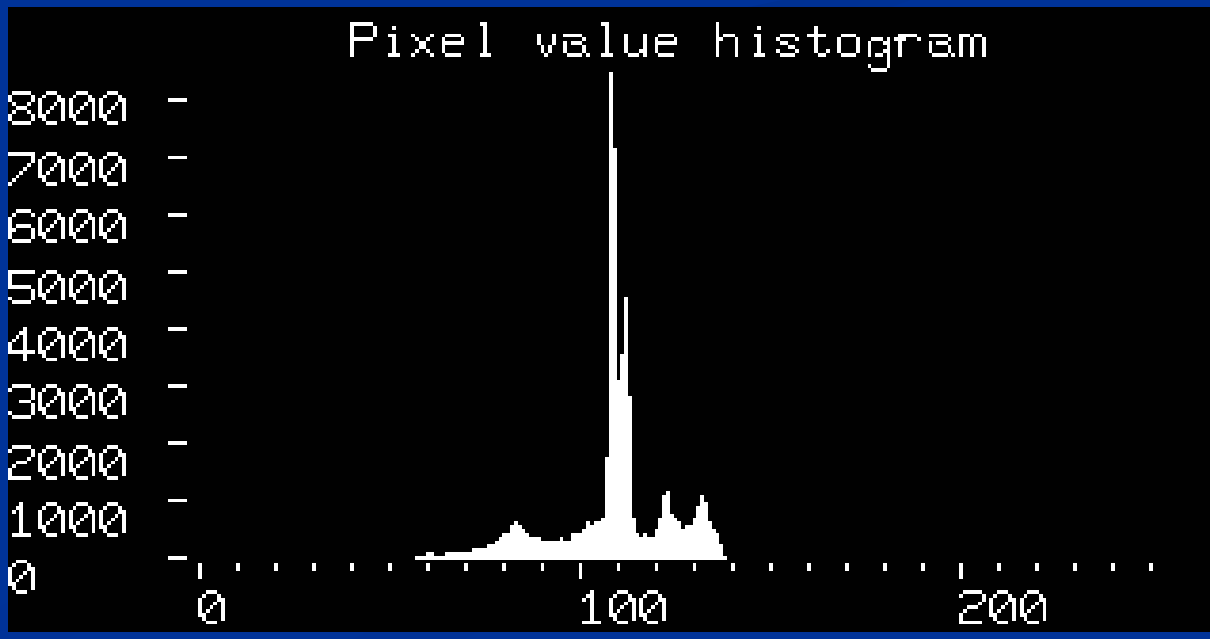
image



histogram

Histogram II

An image has low contrast when the complete range of possible values is not used. Inspection of the histogram shows this lack of contrast.



Histogram of color images

RGB color can be converted to a gray scale value by

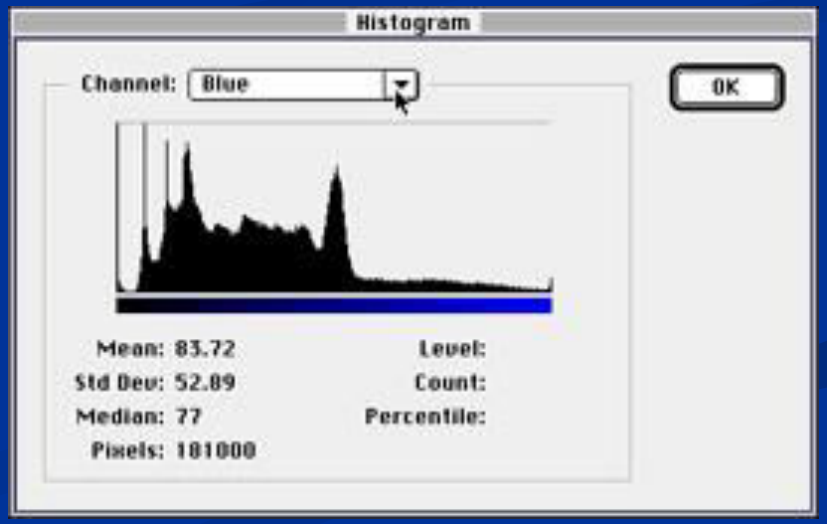
$$Y = 0.299R + 0.587G + 0.114B$$

Y: the grayscale component in the YIQ color space used in NTSC television.
The weights reflect the eye's brightness sensitivity to the color primaries.

Histogram of color images II

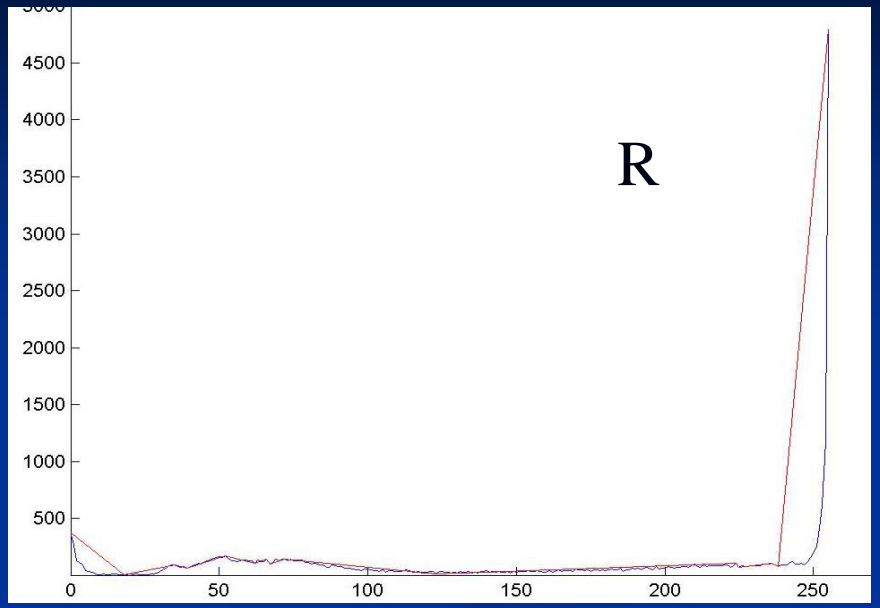
Histogram:

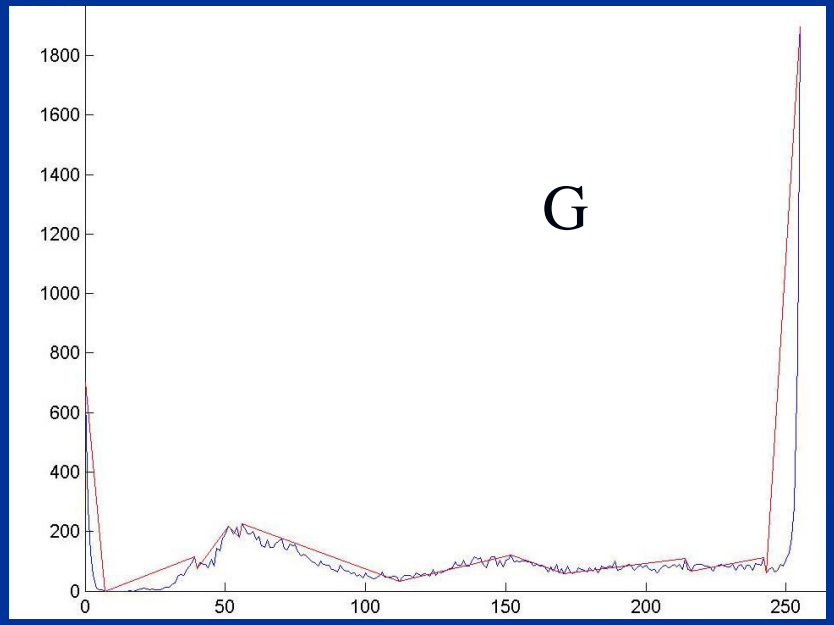
individual histograms of red, green and blue

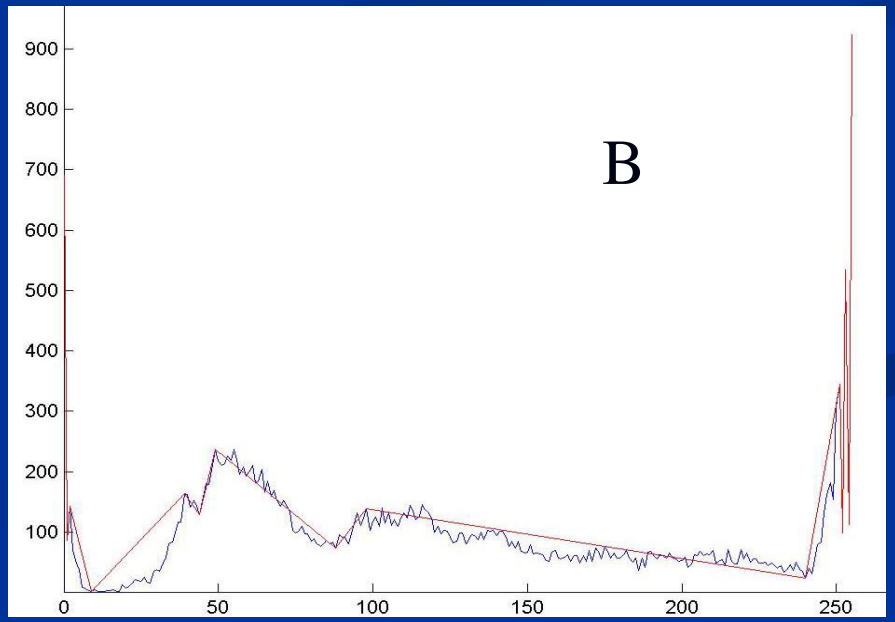


Blue

Histogram of color images III







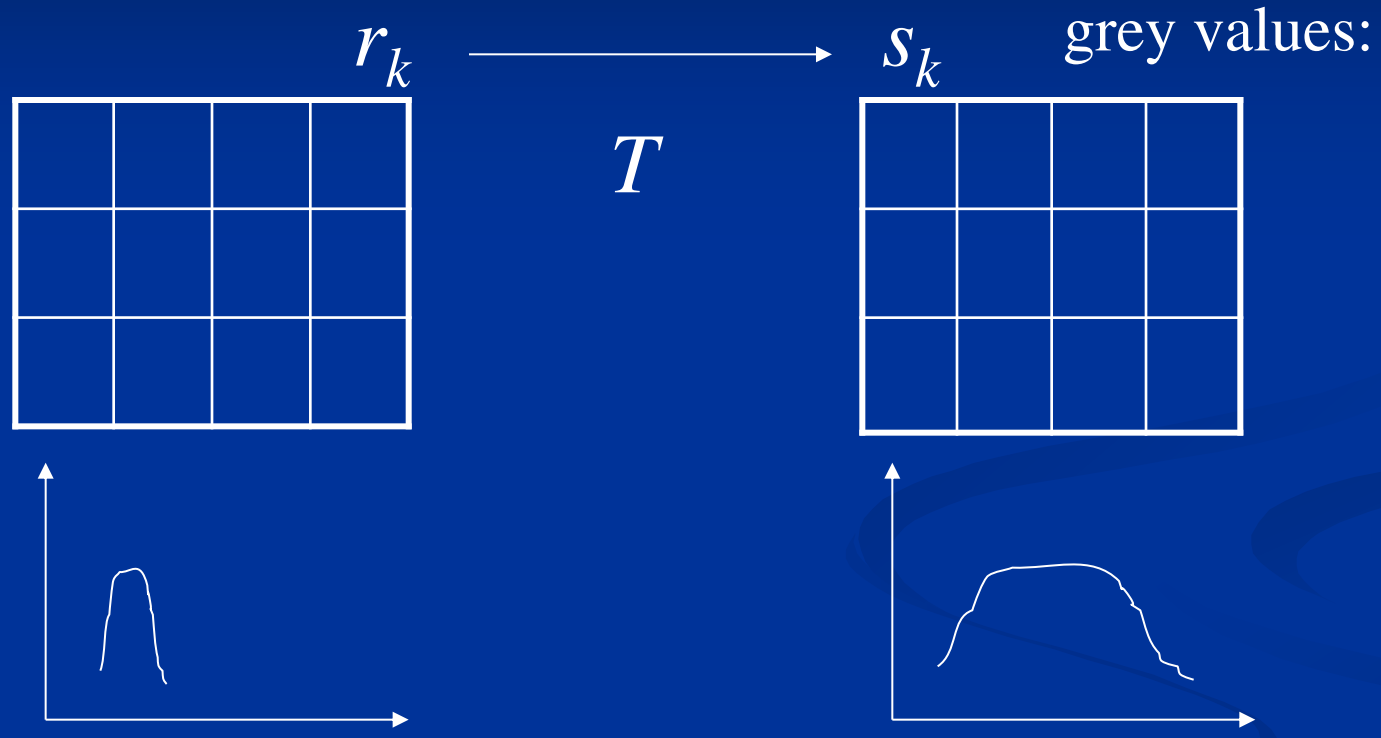
Histogram of color images IV

or

a 3-D histogram can be produced, with the three axes representing the red, blue and green channels, and brightness at each point representing the pixel count

Histogram transformation

Point operation $T(r_k) = s_k$



Properties of T:

keeps the original range of grey values

monoton increasing

Histogram equalization (HE)

transforms the intensity values so that the histogram of the output image approximately matches the flat (uniform) histogram

Histogram equalization II

As for the discrete case the following formula applies:

$$s_k = T(r_k) = \sum_{j=0}^{k} \frac{n_j}{n} \cdot (L\text{-}1)$$

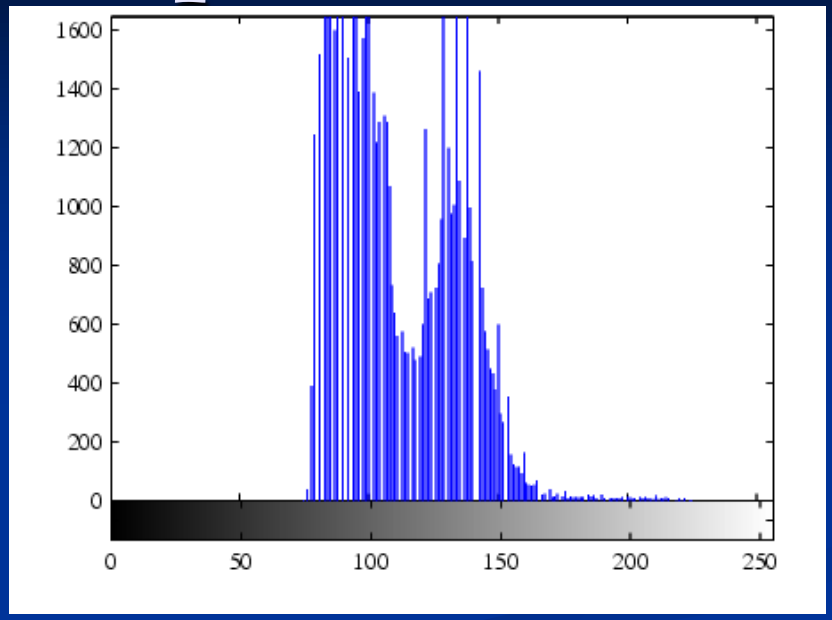
$k = 0,1,2,\ldots,L\text{-}1$

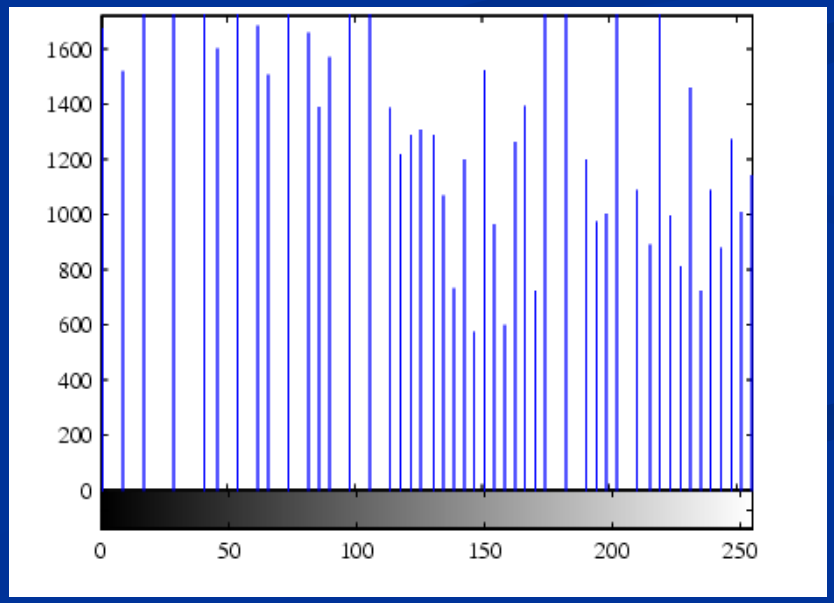
L: number of grey levels in image (e.g., 255)

n_j: number of times j-th grey level appears in image

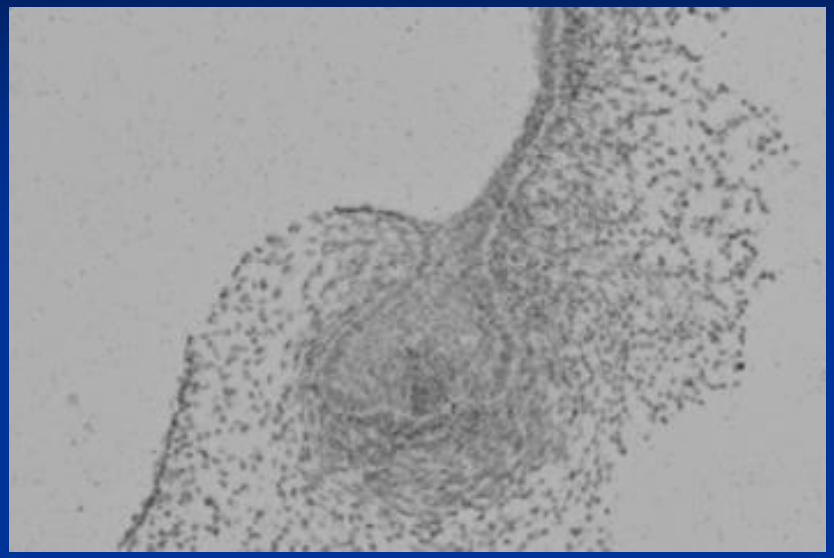
n: total number of pixels in the image

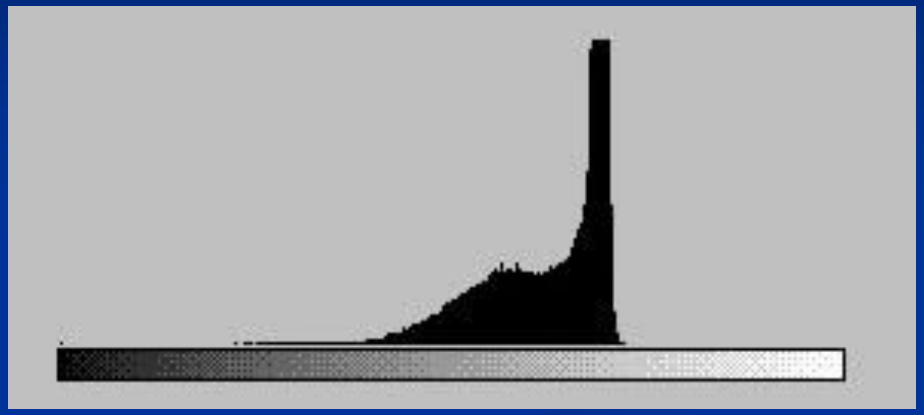
Histogram equalization III

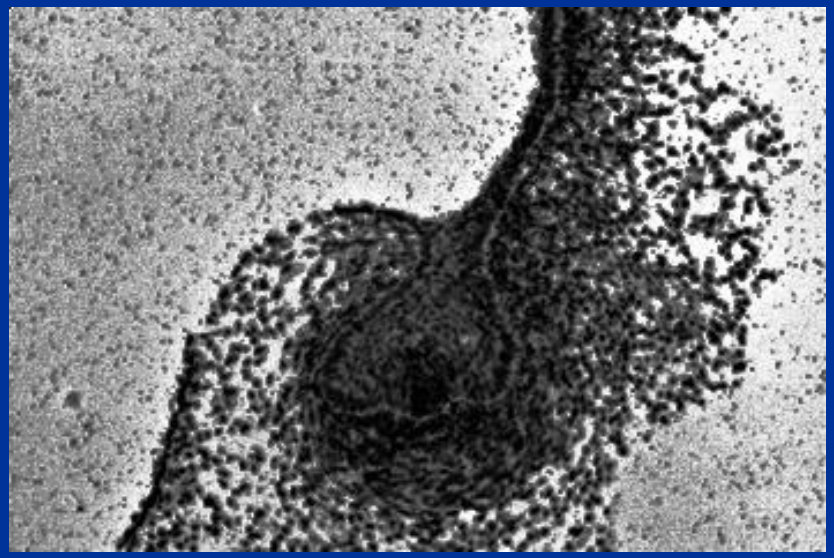


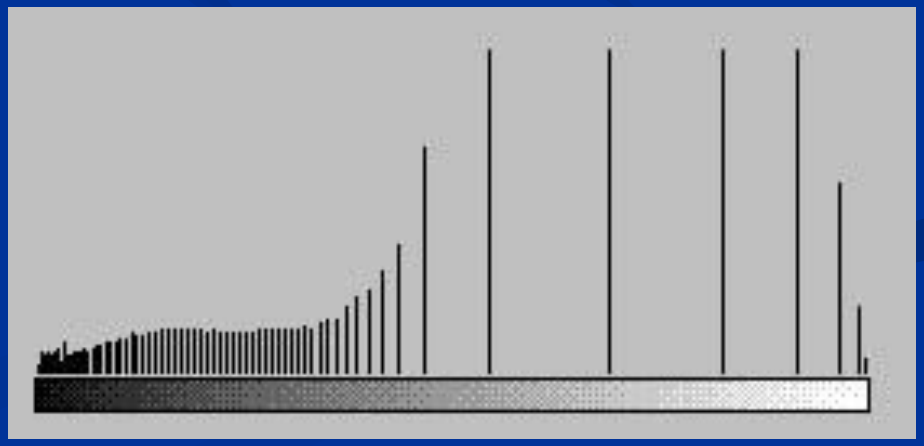


Histogram equalization IV

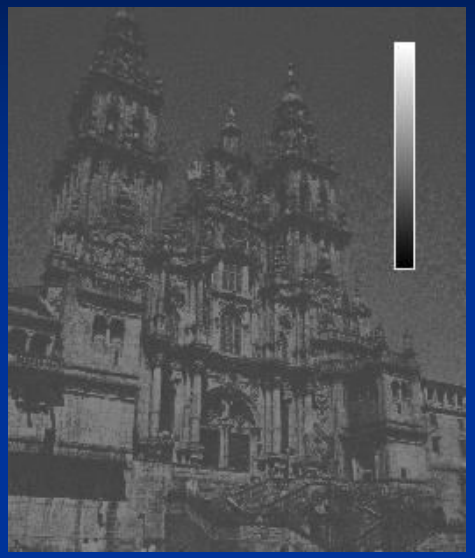


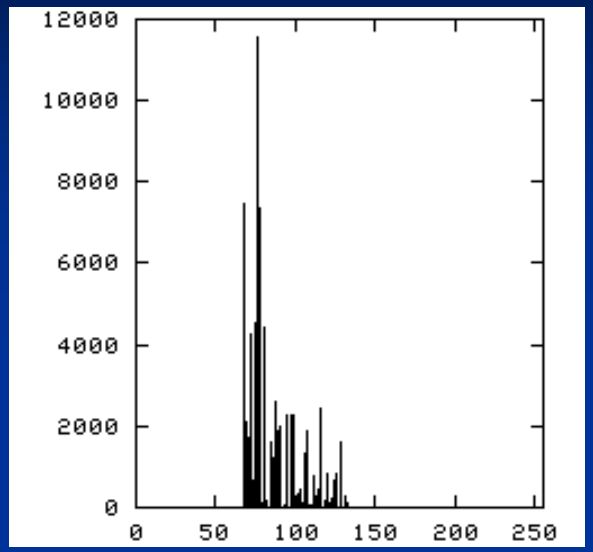


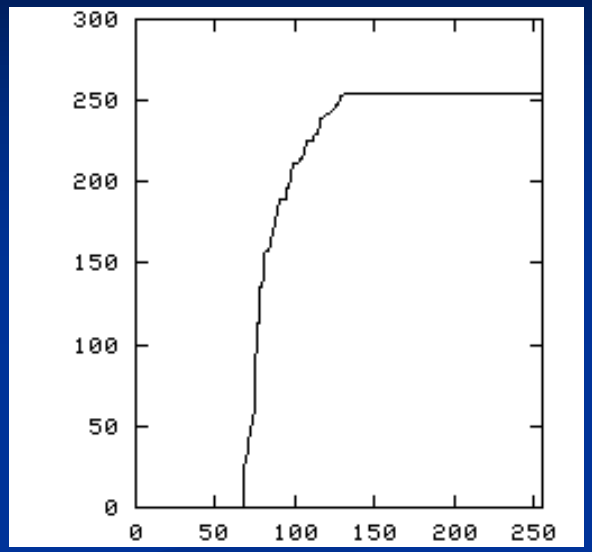




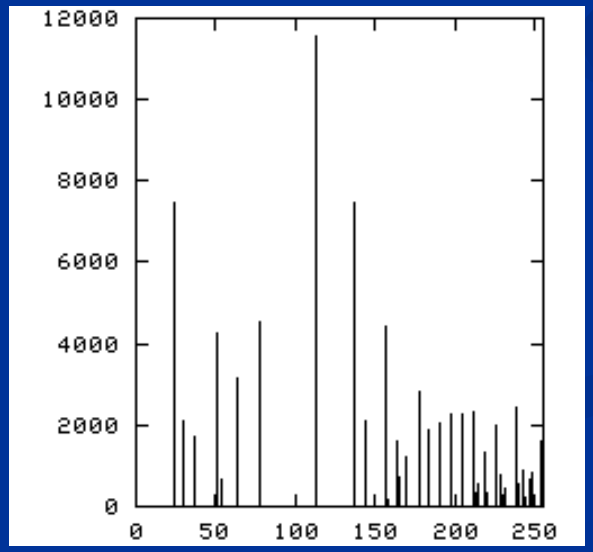
Histogram equalization V



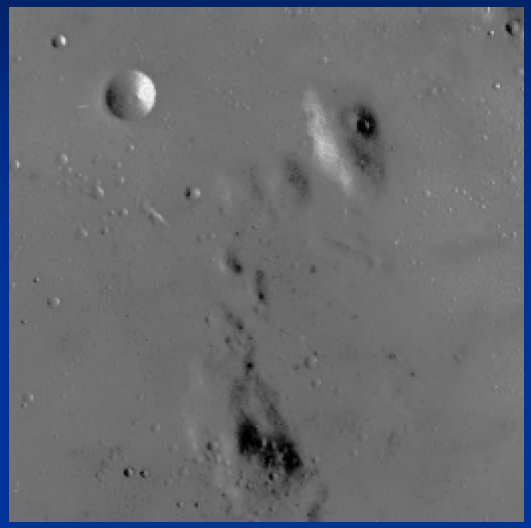


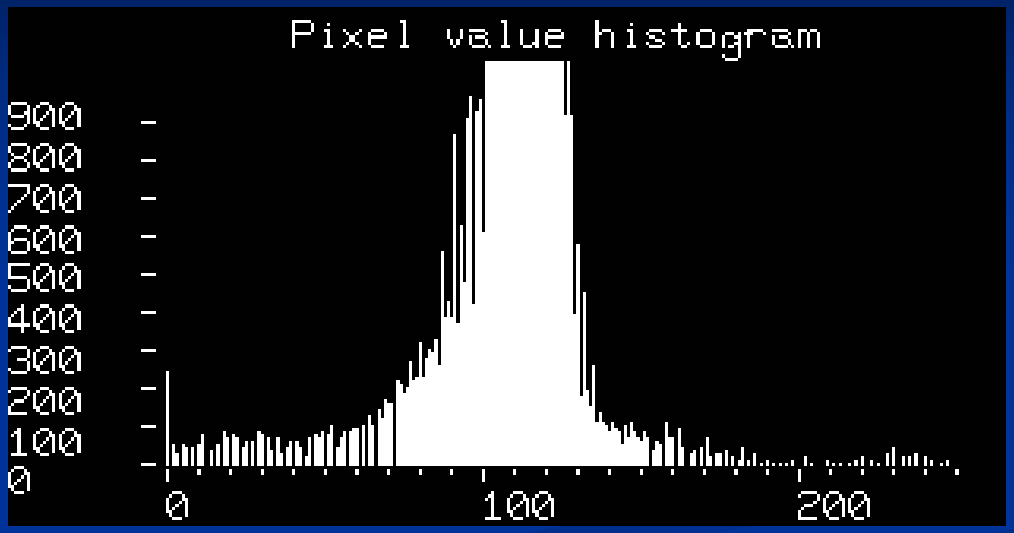


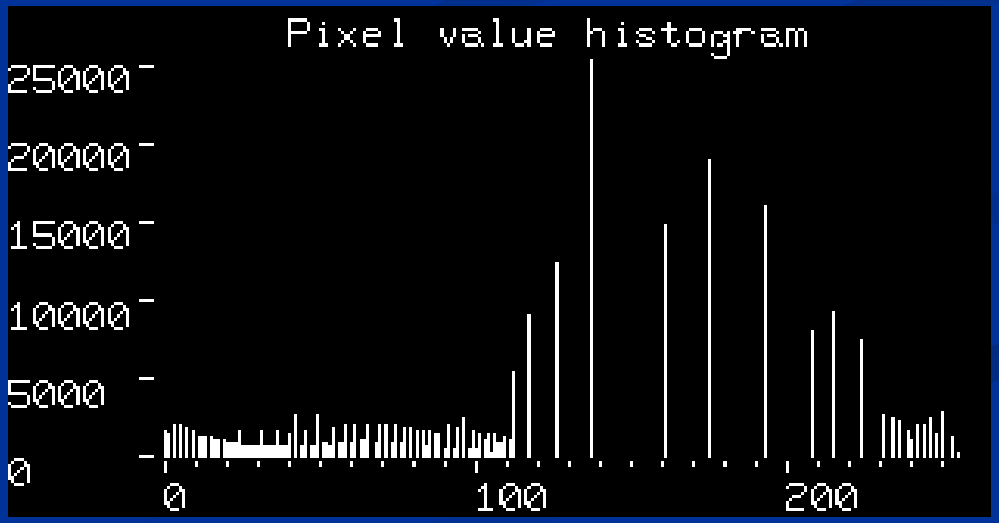
cumulative histogram



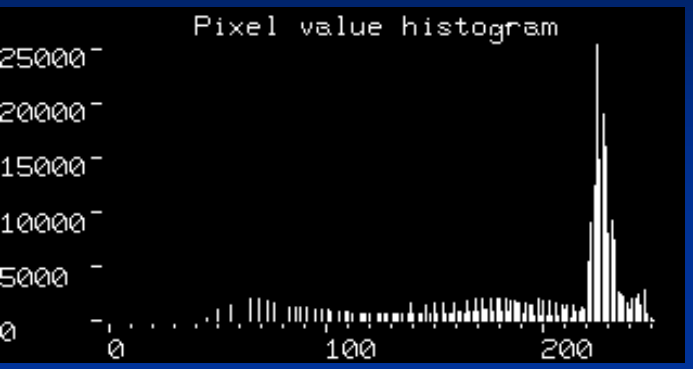
Histogram equalization VI

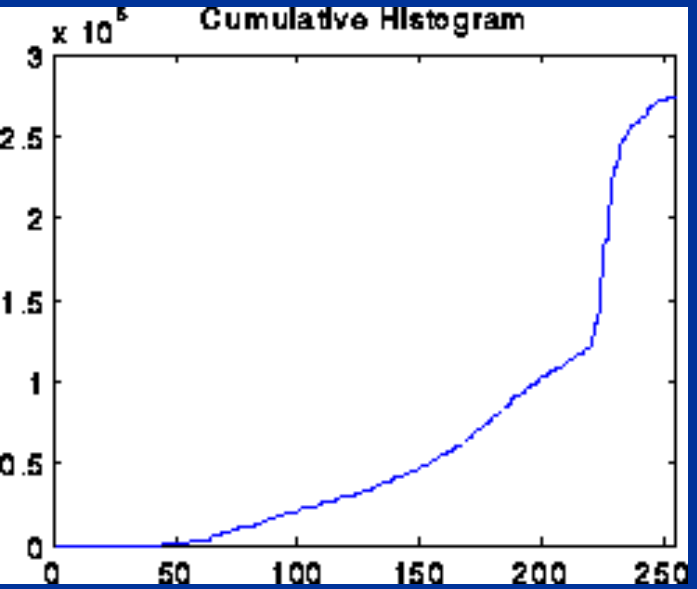






Histogram equalization VII

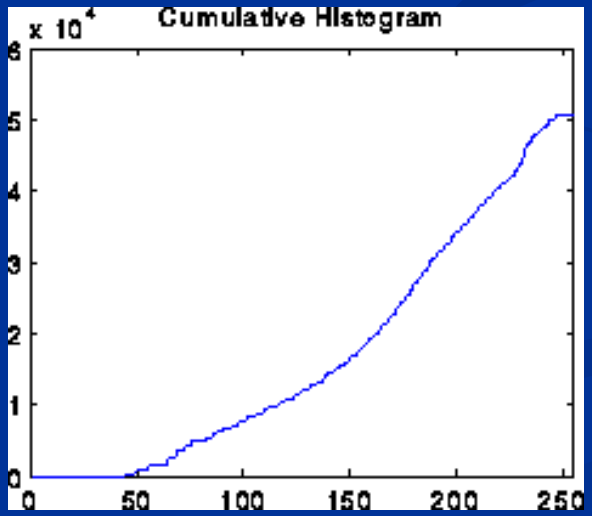




HE

Histogram equalization VIII

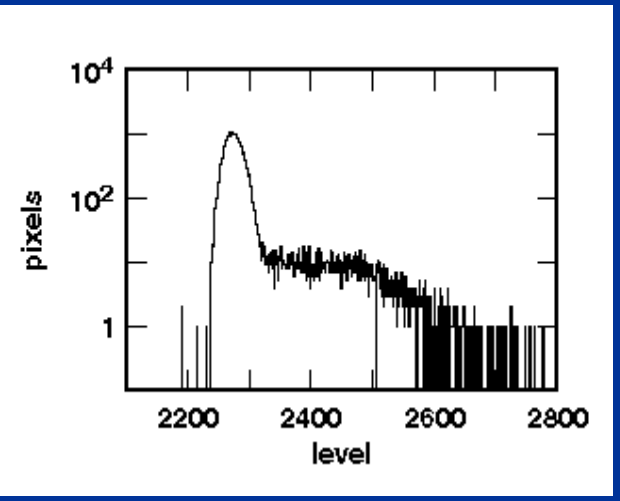
histogram can be taken also on a part of the image

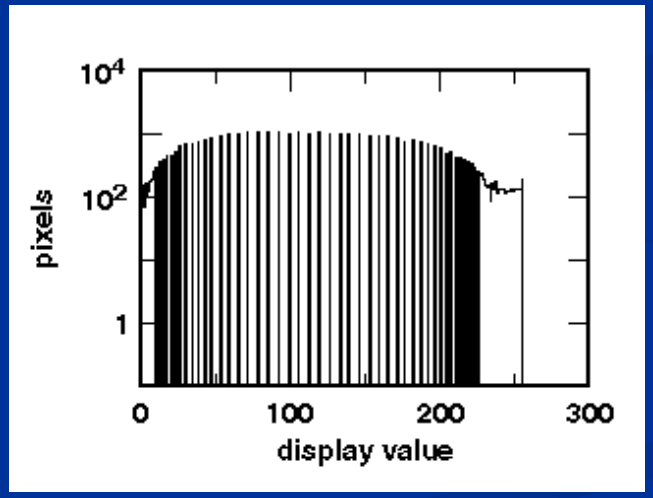


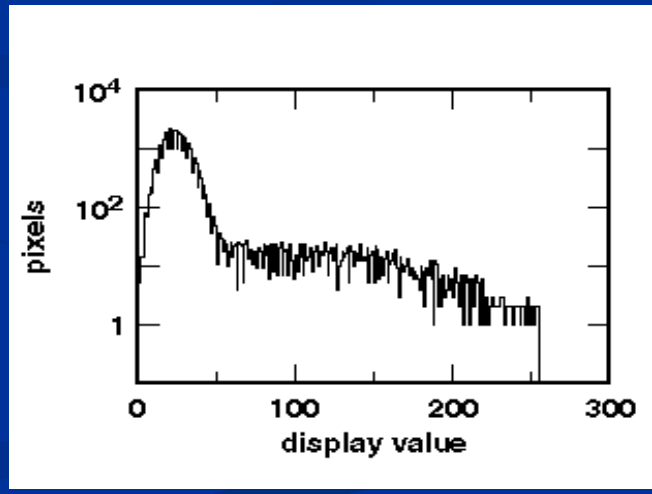
Histogram projection (HP)

assigns equal display space to every occupied raw signal level, regardless of how many pixels are at that same level. In effect, the raw signal histogram is "projected" into a similar-looking display histogram.

Histogram projection II







HE

HP

Histogram projection III

occupied (*used*) *grey level*: there is at least one pixel with that grey level

$B(k)$: the *fraction of* occupied grey levels at or below grey level k
$B(k)$ rises from 0 to 1 in discrete uniform steps of $1/n$, where n is the total number of occupied levels

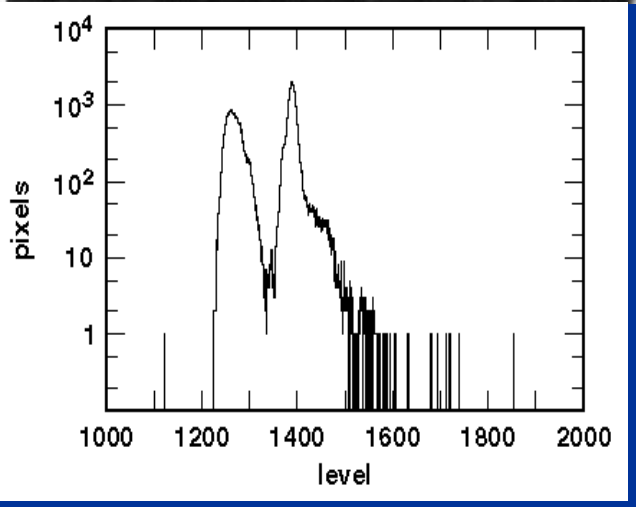
HP transformation:

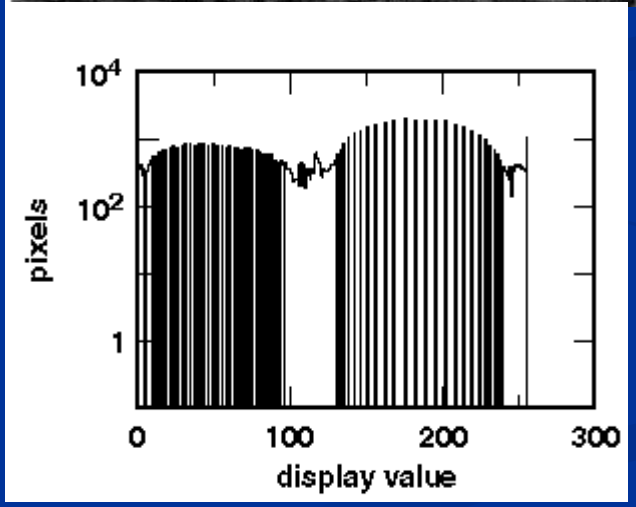
$$s_k = 255 \cdot B(k).$$

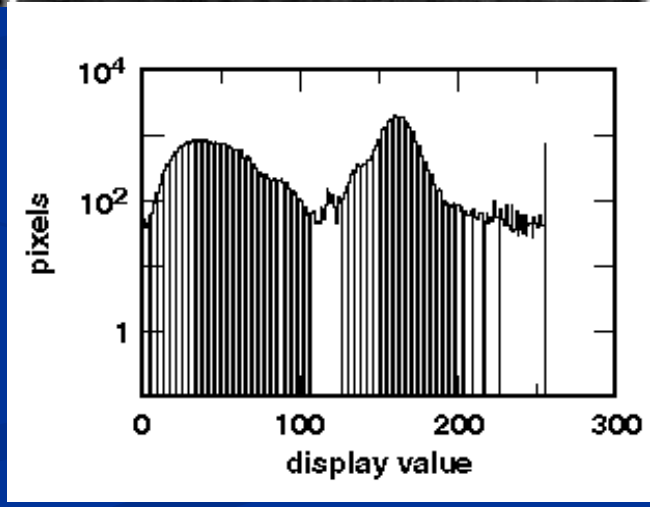
Plateau equalization

By clipping the histogram count at a saturation or *plateau value*, one can produce display allocations intermediate in character between those of HP and HE.

Plateau equalization II







HE

PE 50

Plateau equalization III

The PE algorithm computes the distribution not for the full image histogram but for the histogram clipped at a plateau (or saturation) value in the count.
When that plateau value is set at 1, we generate $B(k)$ and so perform HP;
When it is set above the histogram peak, we generate F(k) and so perform HE.
At intermediate values, we generate an intermediate distribution which we denote by $P(k)$.

PE transformation:

$$s_k = 255 \cdot P(k)$$

Histogram specification (HS)

an image's histogram is transformed according to a desired function

Transforming the intensity values so that the histogram of the output image approximately matches a specified histogram.

Histogram specification II

IMAGE

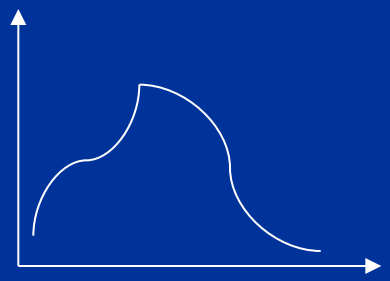
DESIRED IMAGE

histogram$_1$

T

S

histogram$_2$



$S^{-1} * T$

Contrast streching (CS)

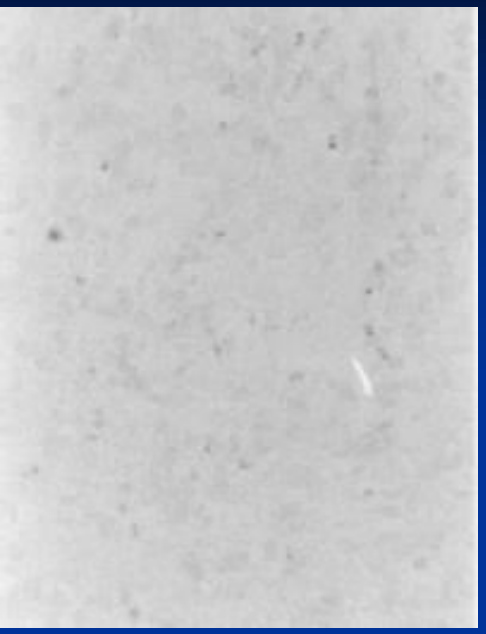
By stretching the histogram we attempt to use the available full grey level range.

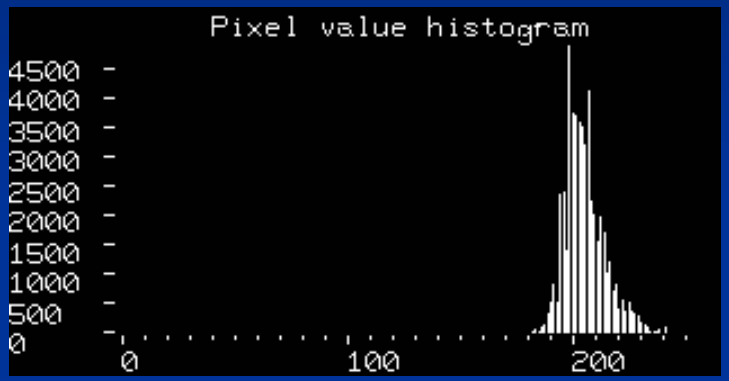
The appropriate CS transformation :

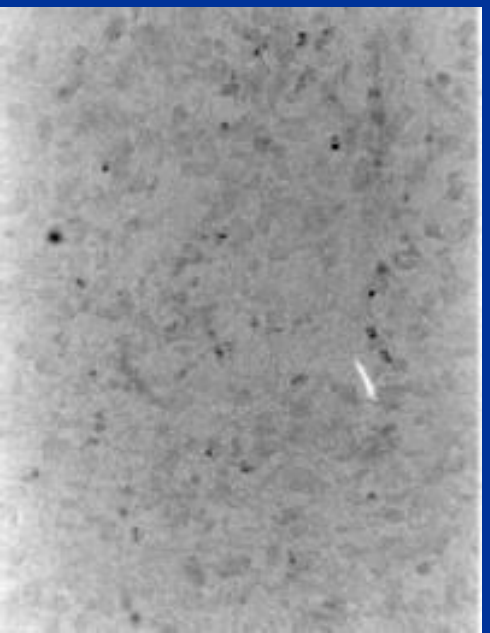
$$s_k = 255 \cdot (r_k\text{-min})/(\text{max-min})$$

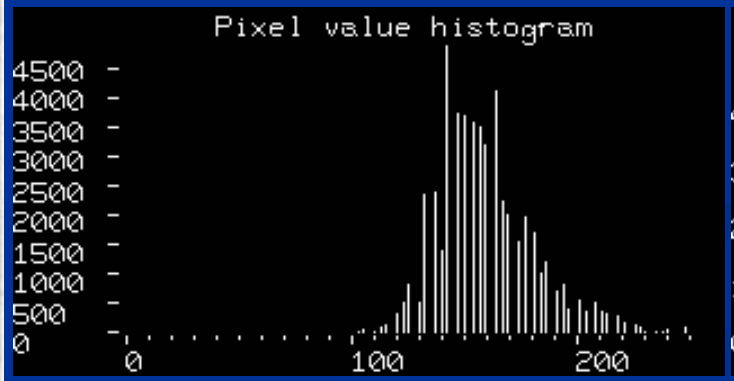
It can also map an intensity range to another intensity range.

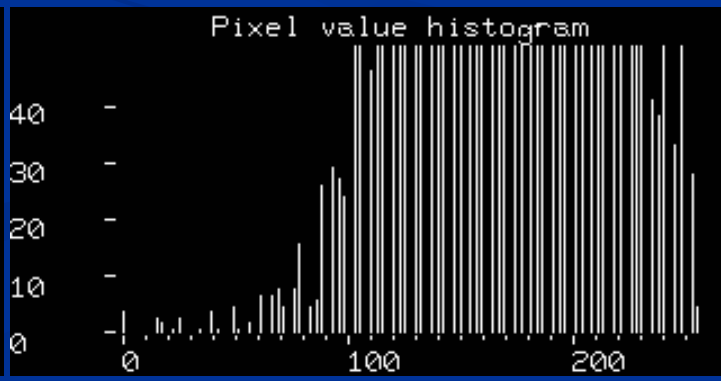
Contrast streching II



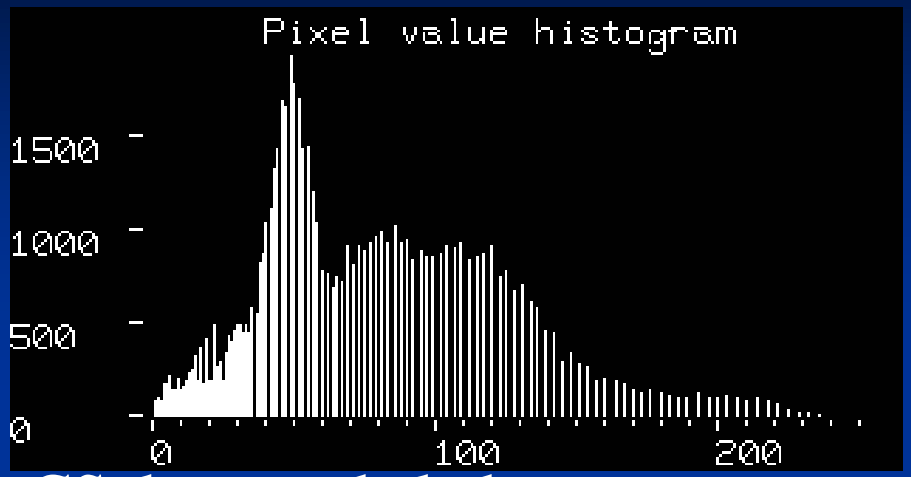




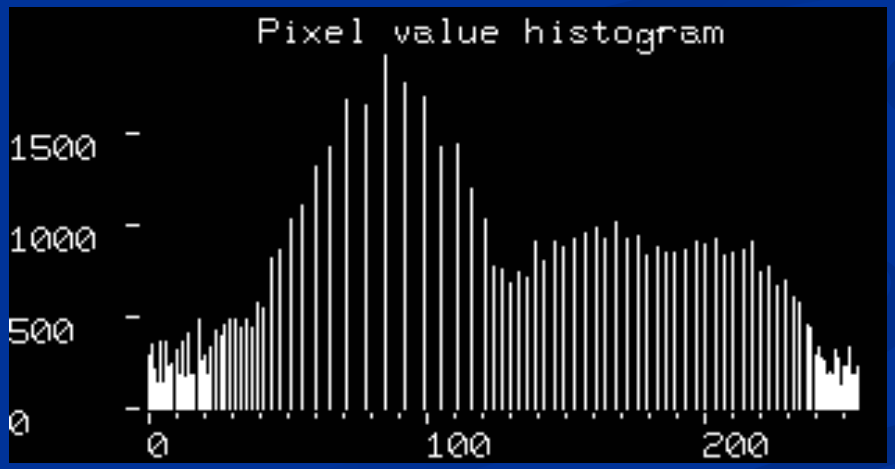




Contrast streching III



CS does not help here



HE

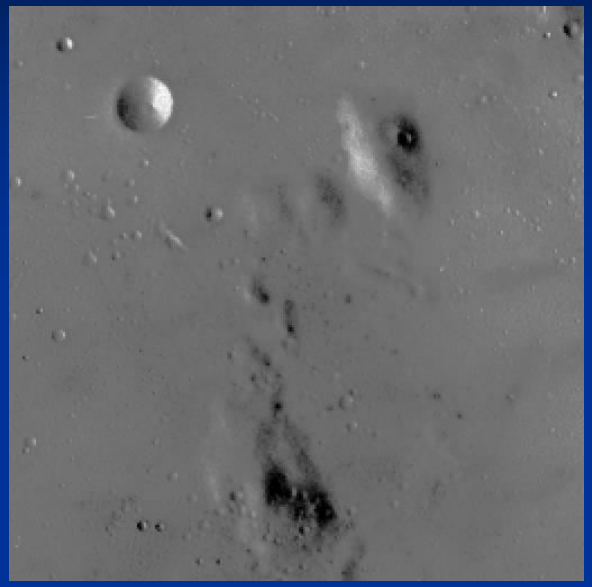
?

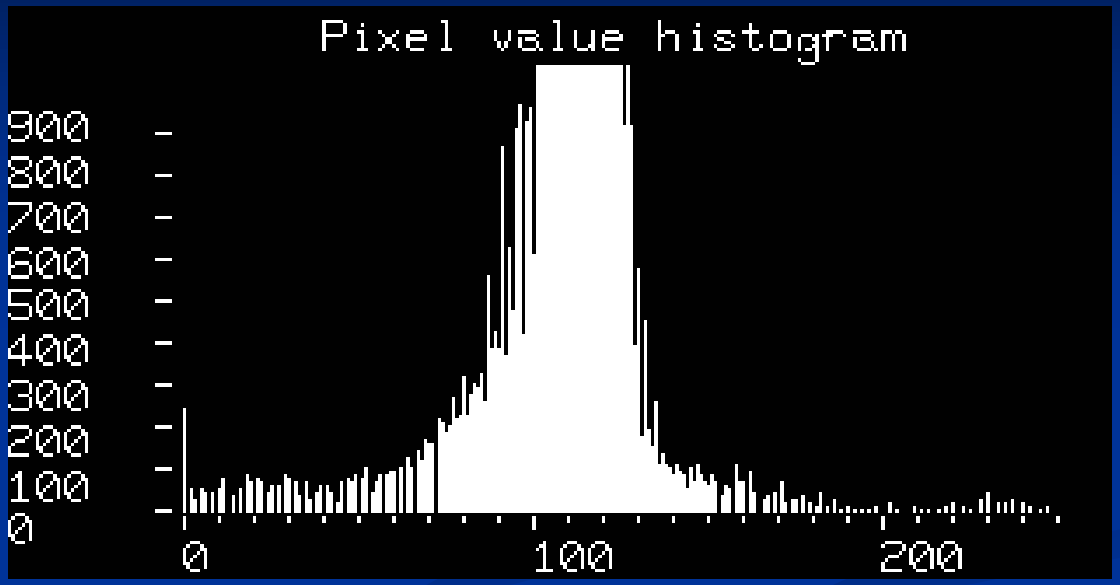
Contrast streching IV

CS

HE

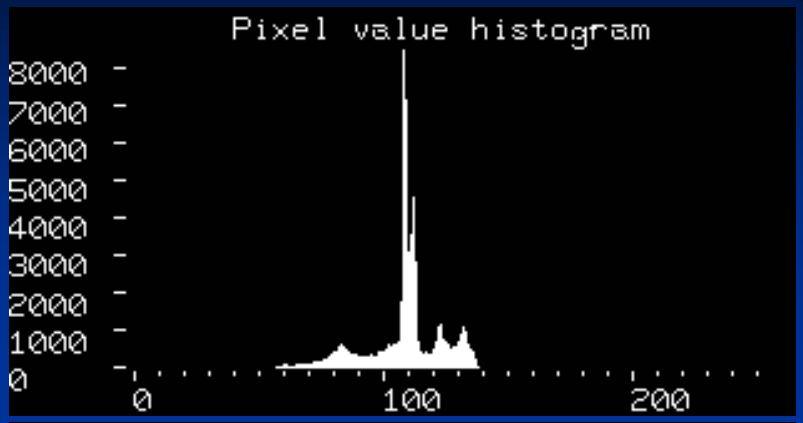
Contrast streching V



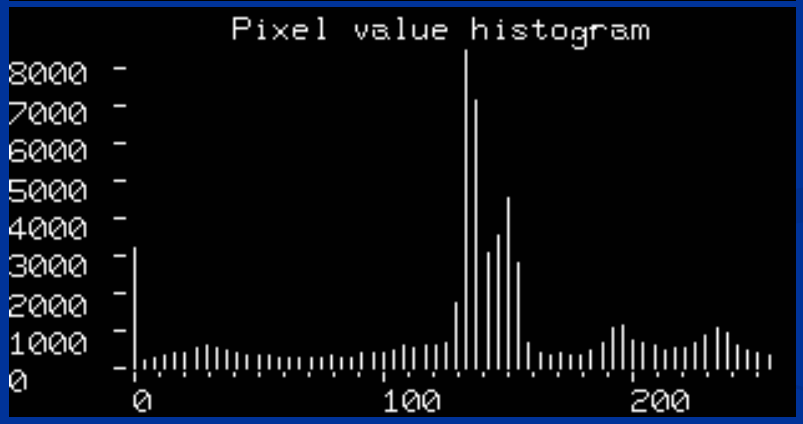


CS
1% - 99%

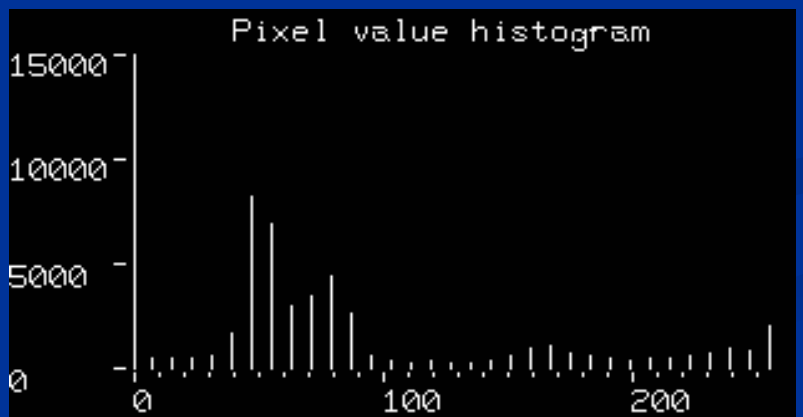
Contrast streching VI



HE

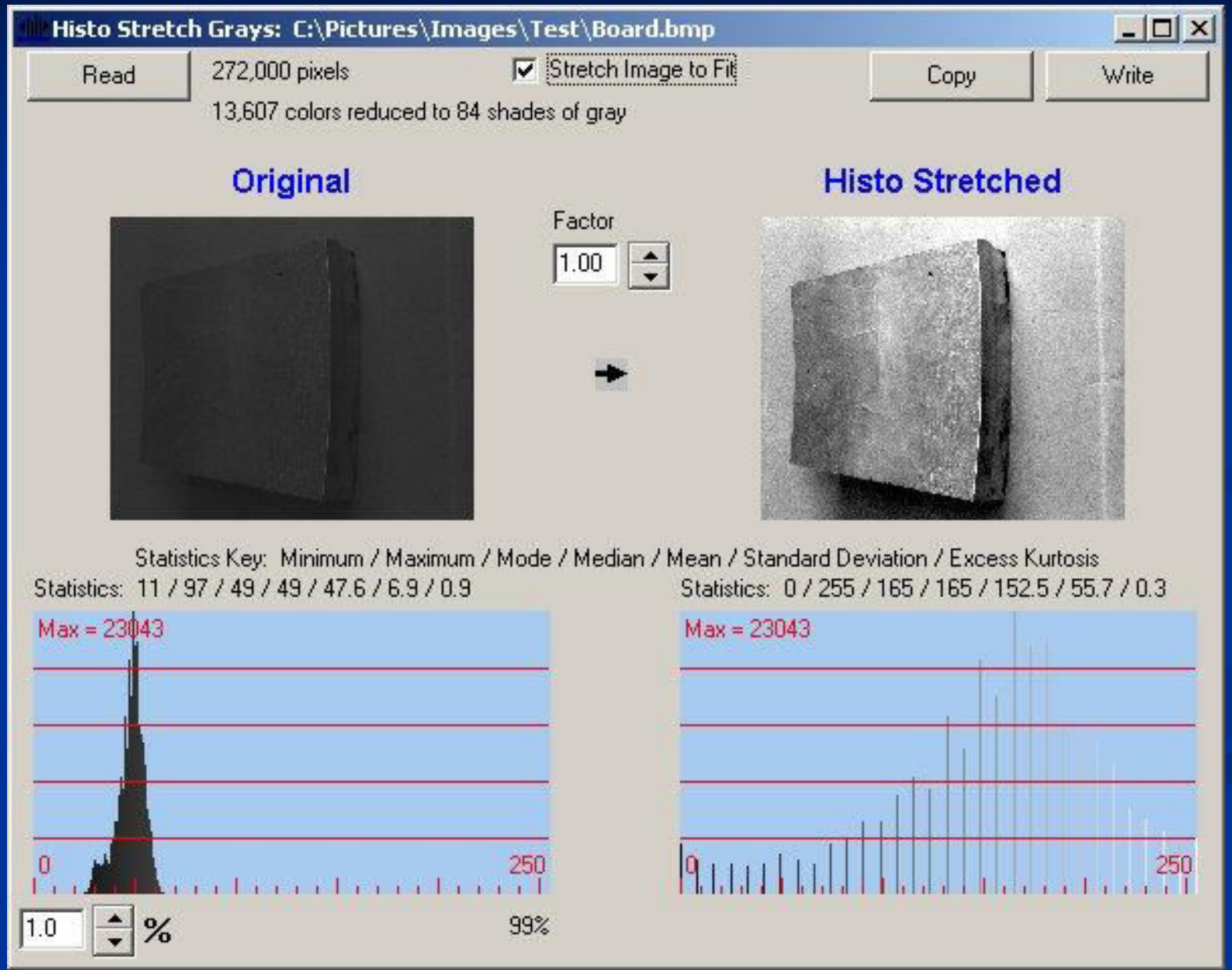


CS
79, 136

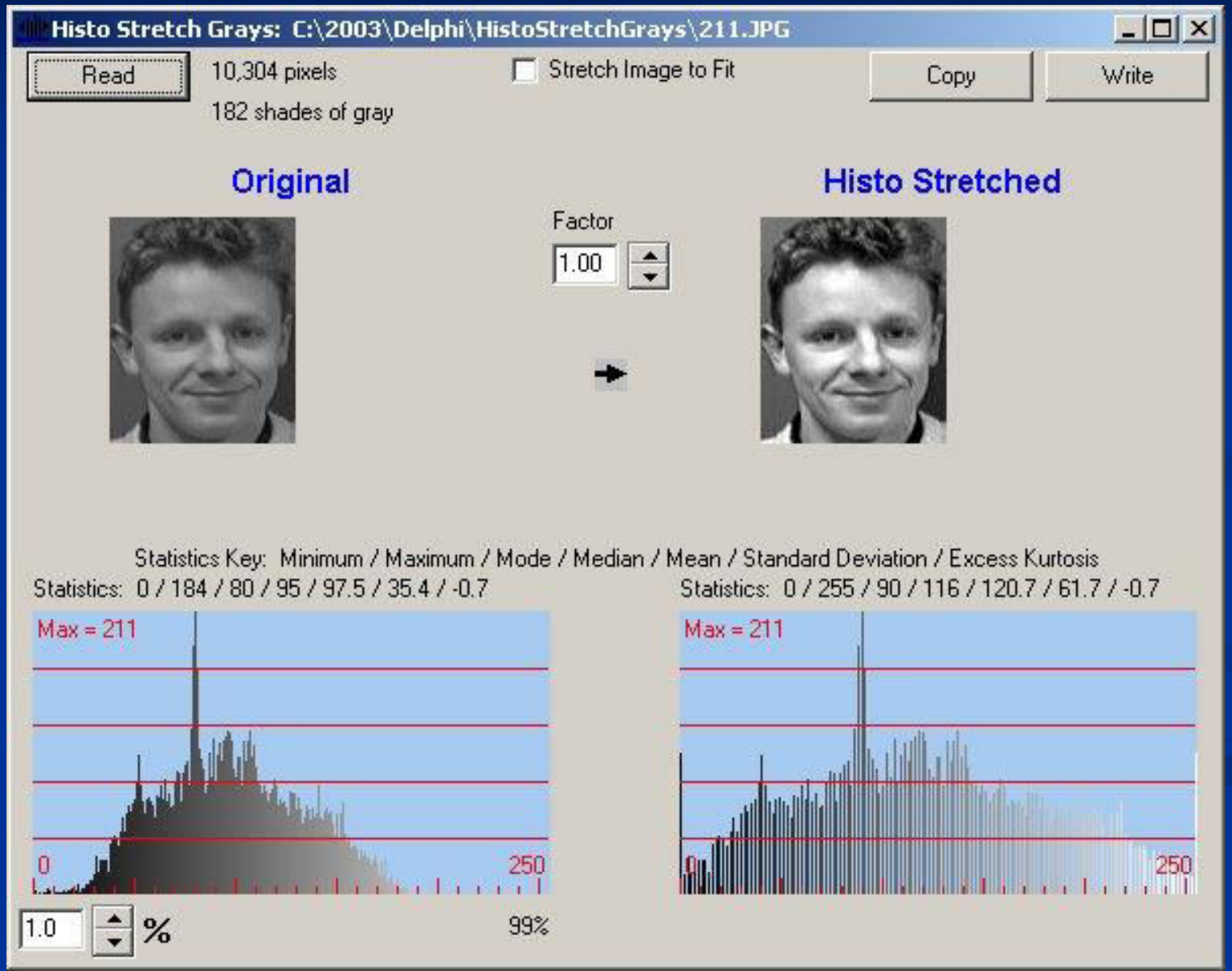


CS
Cutoff fraction: 0.8

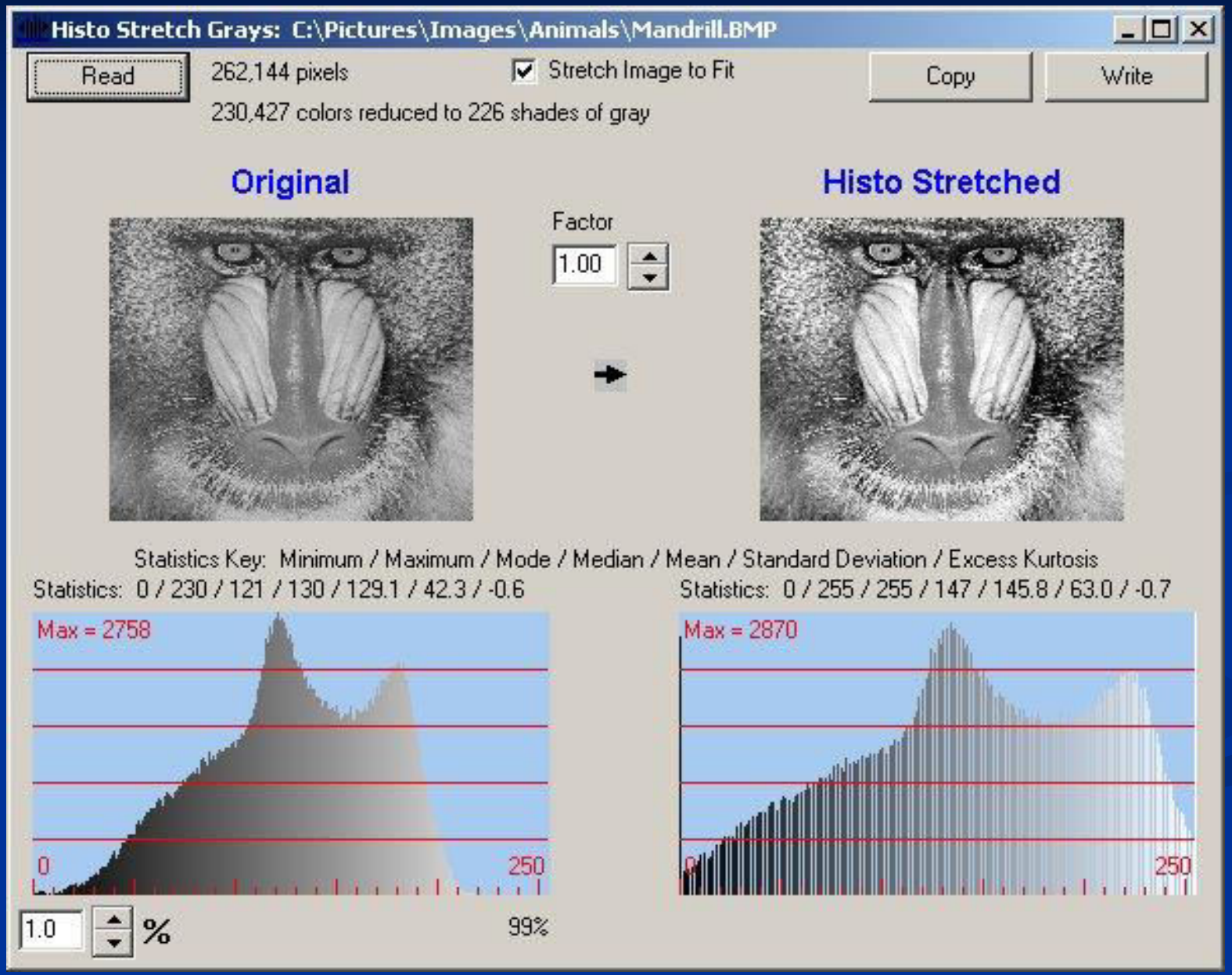
Contrast streching IX



Contrast streching X



Contrast streching XI



Applications

CT lung studies

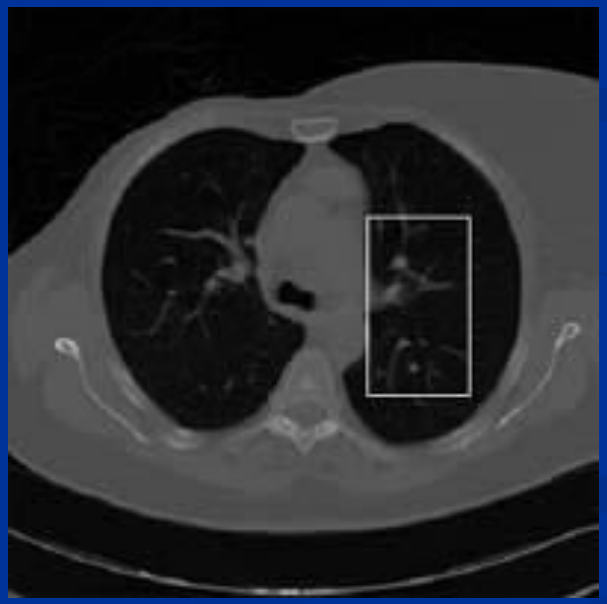
Thresholding

Normalization

Normalization of MRI images

Presentation of high dynamic images (IR, CT)

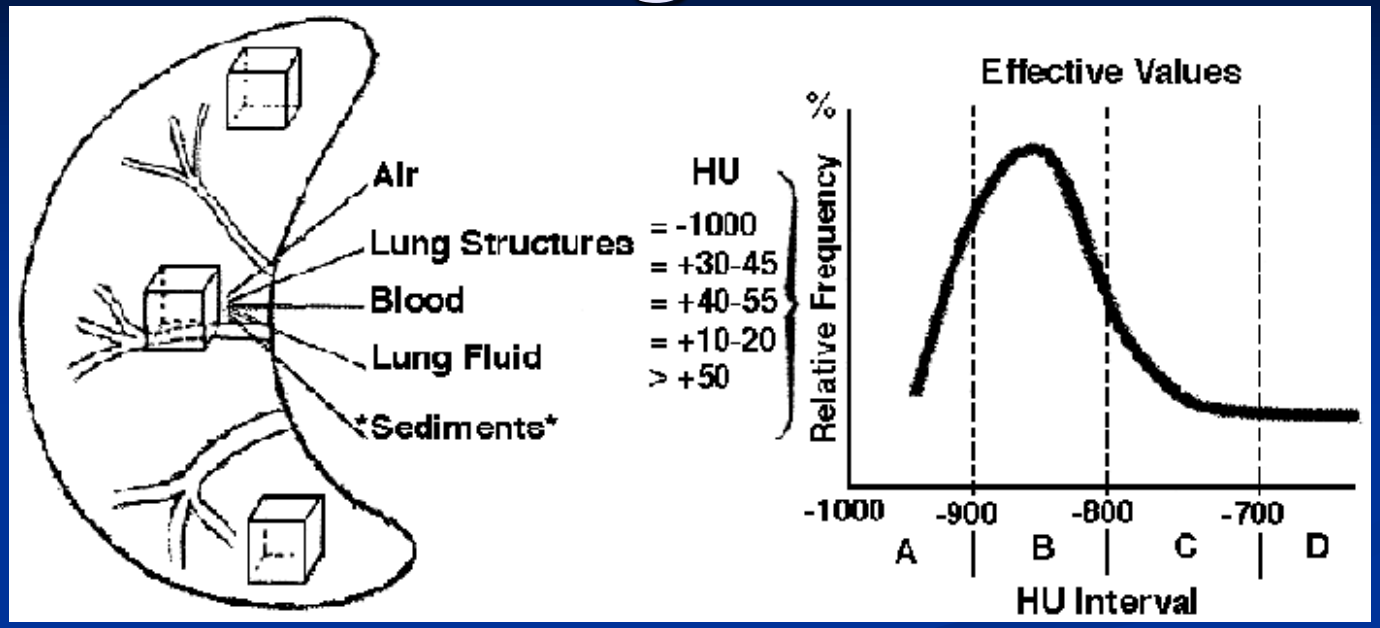
CT lung studies



Yinpeng Jin

HE taken in a part of the image

CT lung studies



Frequency distribution of CT values as percentage (mean +/- SEM) at defines HU intervals in groups of healthy controls and patients with DFA, GLD, and a1-PID

HU interval	Control (n=6)	DFA (n=27)	GLD (n=27)	a1-PID (n=3)
A:less than -900	12.8+/-1.7	11.9+/-2.3	12.6+/-1.9	46.3+/-2.7a
B.-900 to -800	56.7+/-2.1	36.8+/-3.5b	42.5+/-2.9b	33.4+/-2.6a
C: -799 to -700	17.1+/-1.9	22.4+/-2.1	21.6+/-1.8	7.5+/-1.9a
D: greater than -699	14.4+/-0.6	28.9+/-2.4a	23.4+/-3.0a	10.8+/-1.5

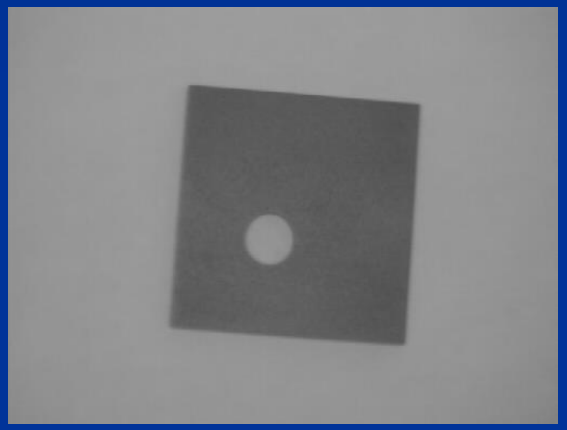
ap<0.05 vs. control. bp<0.01 vs. control.

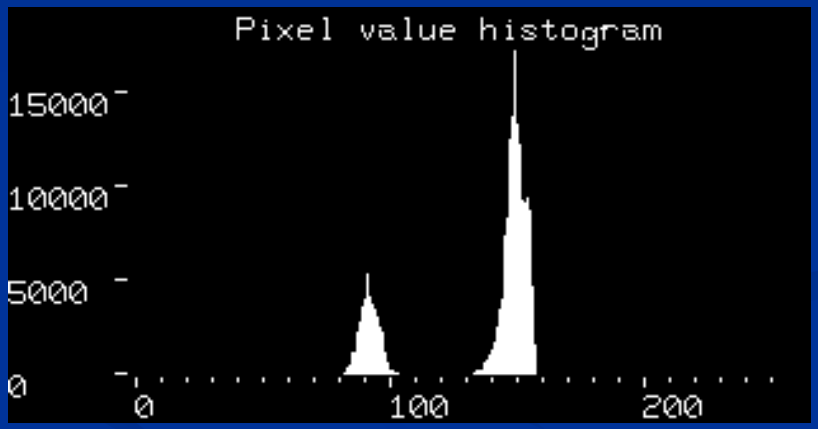
R.Rienmuller

Thresholding

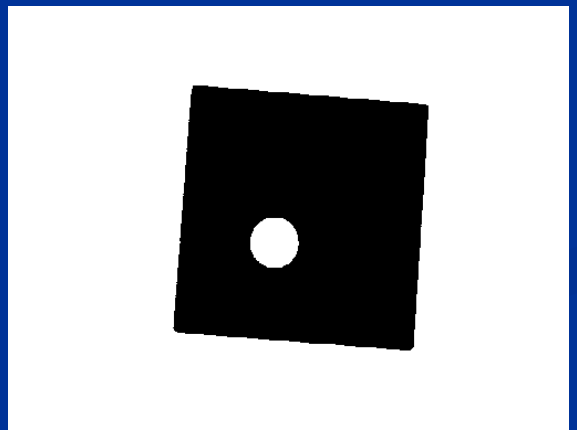
converting a greyscale image to a binary one

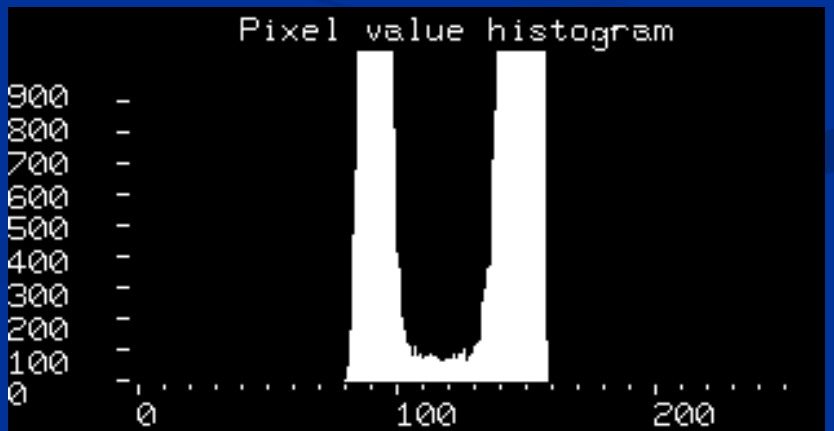
for example, when the histogram is bi-modal





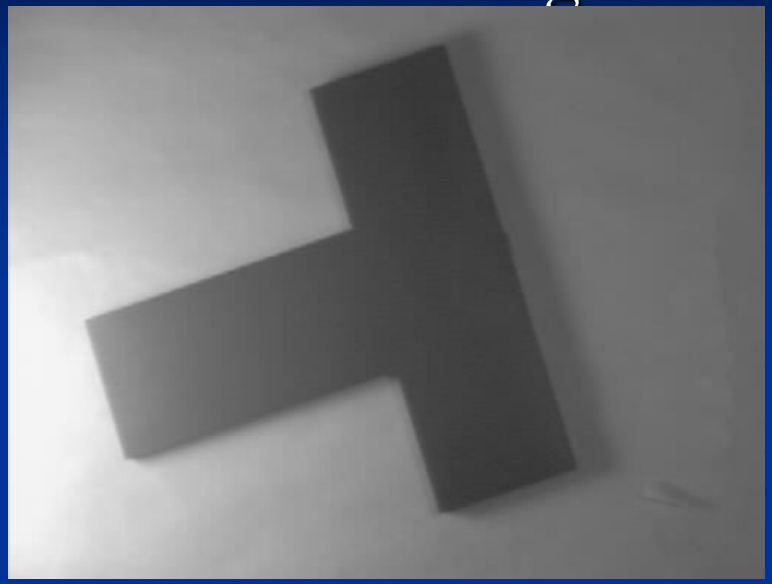
threshold: 120

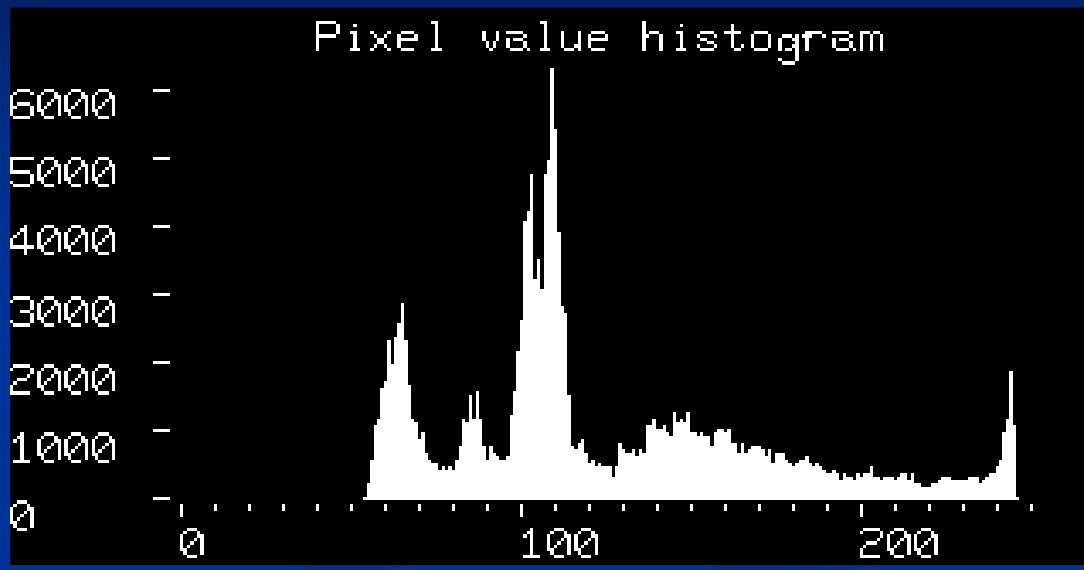


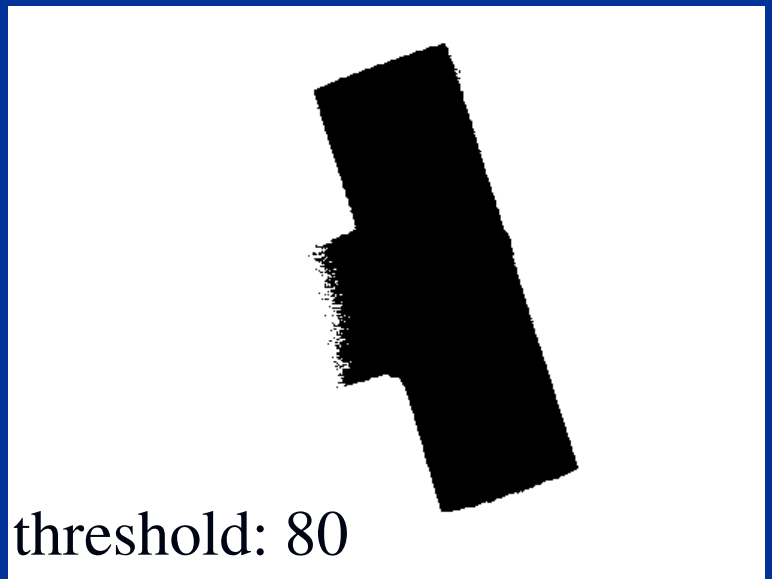


Thresholding II

when the histogram is not bi-modal







threshold: 80

threshold: 120

Normalization I

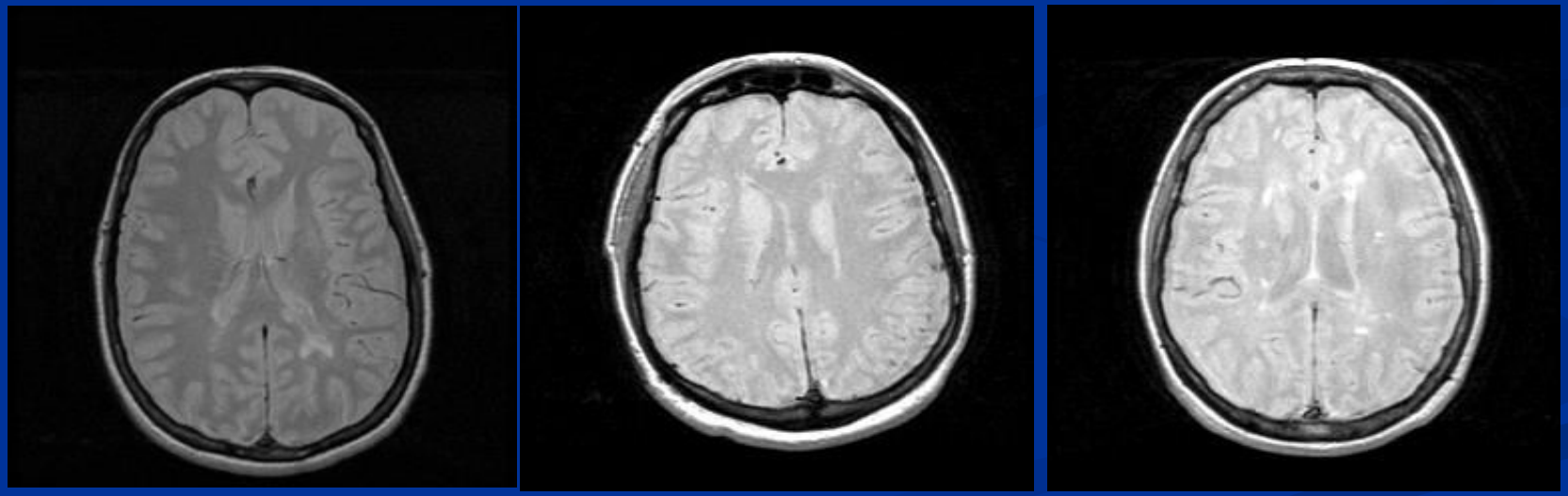
When one wishes to compare two or more images on a specific basis, such as texture, it is common to first normalize their histograms to a "standard" histogram. This can be especially useful when the images have been acquired under different circumstances. Such a normalization is, for example, HE.

Normalization II

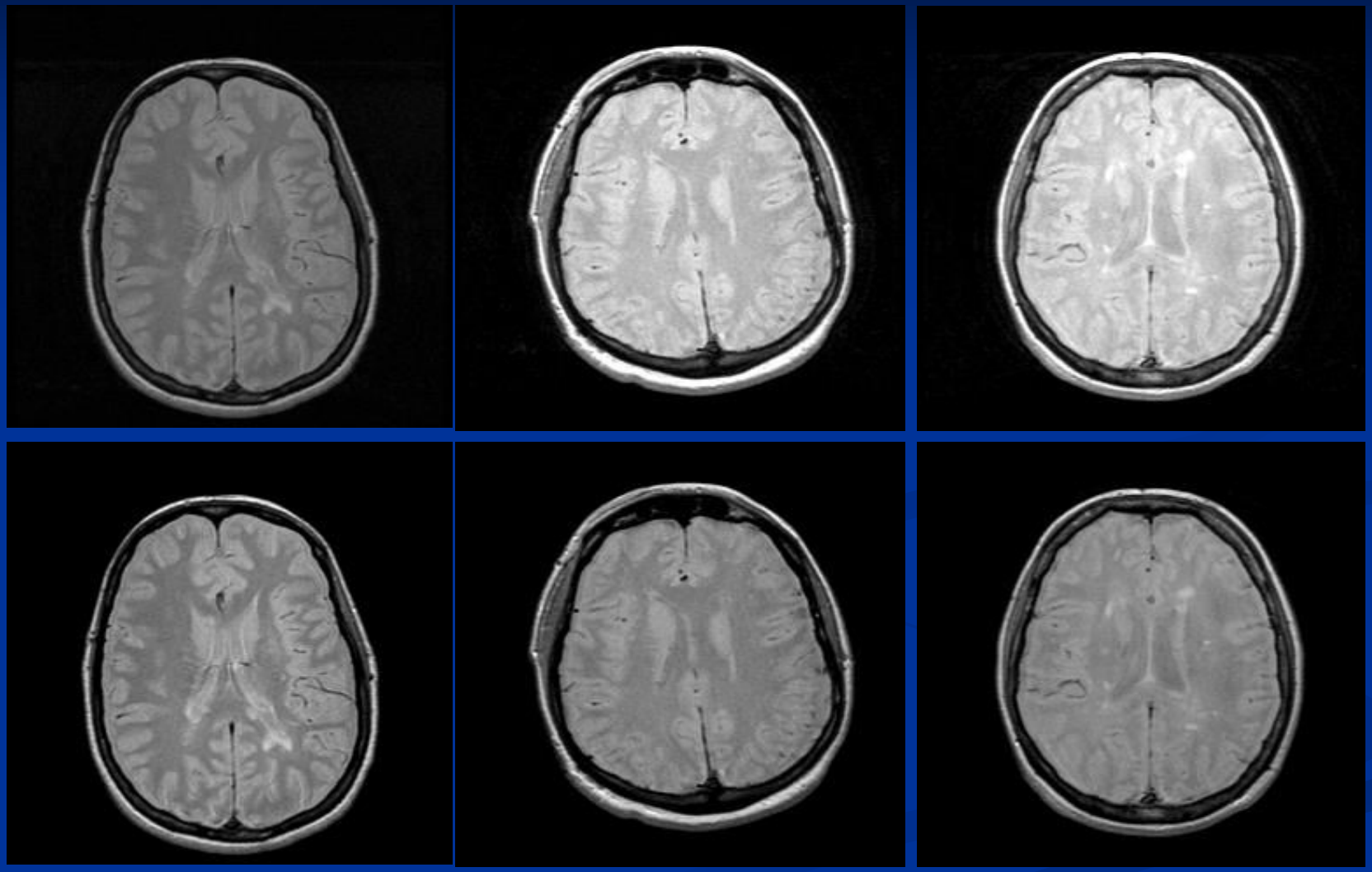
Histogram matching takes into account the shape of the histogram of the original image and the one being matched.

Normalization of MRI images

MRI intensities do not have a fixed meaning, not even within the same protocol for the same body region obtained on the same scanner for the same patient.



Normalization of MRI images II



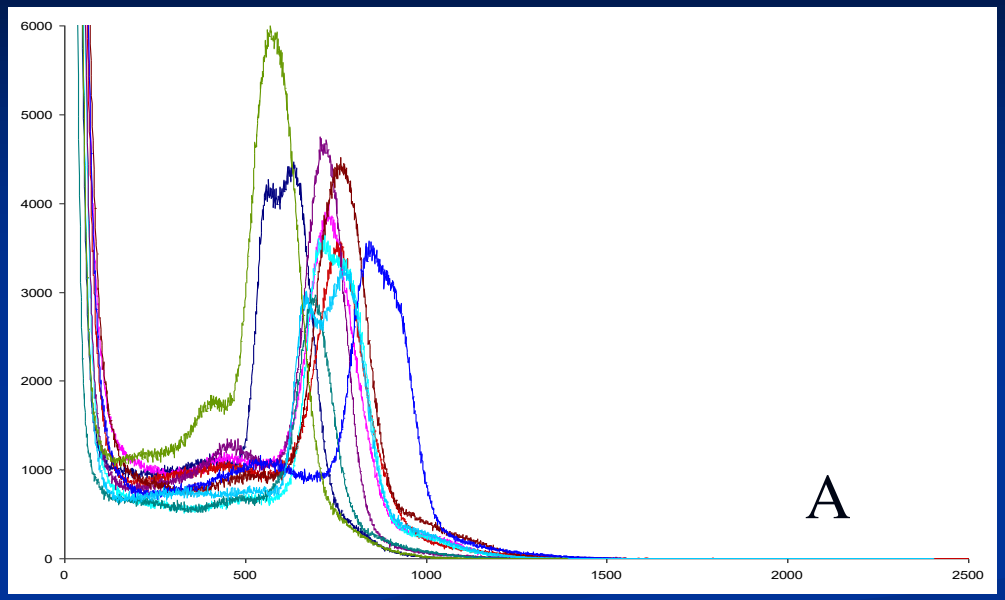
L. G. Nyúl, J. K. Udupa

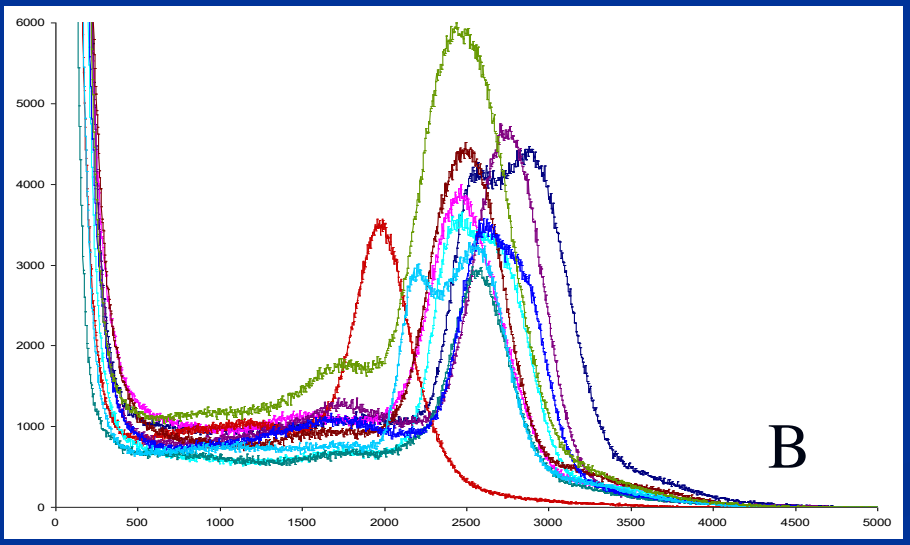
Normalization of MRI images III

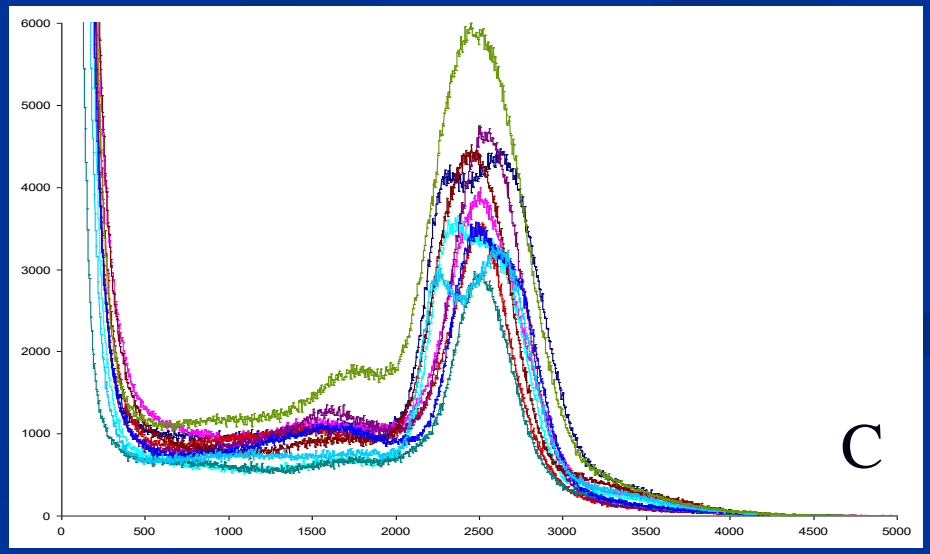
A: Histograms of 10 FSE PD brain volume images of MS patients.

B: The same histograms after scaling.

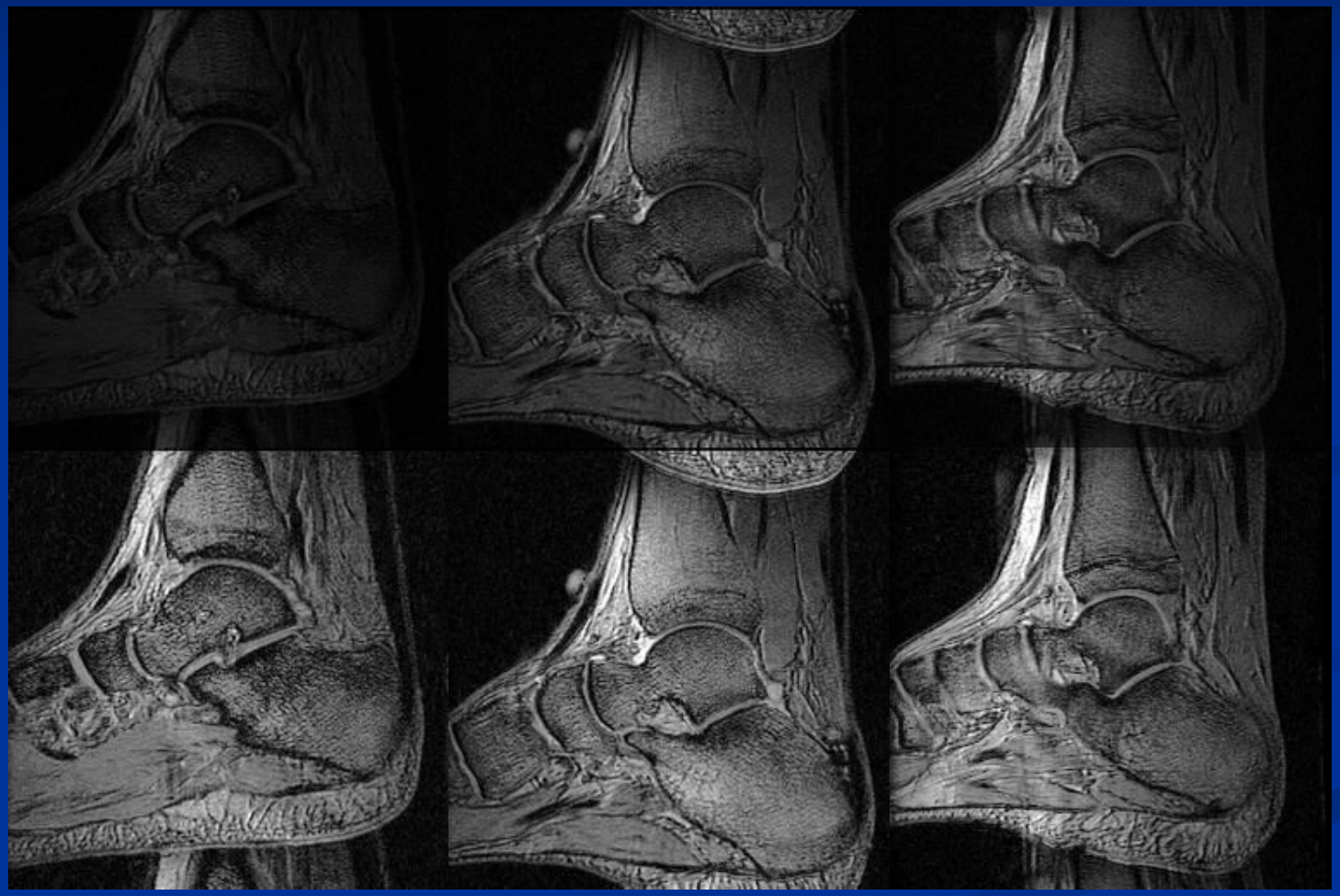
C: The histograms after final standardization.







Normalization of MRI images V



Applications III

A digitized high dynamic range image, such as an infrared (IR) image or a CAT scan image, spans a much larger range of levels than the typical values (0 - 255) available for monitor display. The function of a good display algorithm is to map these digitized raw signal levels into display values from 0 to 255 (black to white), preserving as much information as possible for the purposes of the human observer.

Applications IV

The HP algorithm is widely used by infrared (IR) camera manufacturers as a real-time automated image display.

The PE algorithm is used in the B-52 IR navigation and targeting sensor.